

Bharath Prakash

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Professional Summary

Robotics software engineer with hands-on experience building real-time ROS 2 control stacks, hardware interfaces, and motion-planning pipelines for industrial cobots and robots, alongside full-stack systems engineering across desktop applications, concurrent data pipelines, and embedded computer-vision prototypes. Pursuing an M.Sc. in Communication and Media Engineering at Hochschule Offenburg.

Professional Experience

Work-Life Robotics Institute

Mar 2026 – Present

Student Research Assistant - Robotics Software Engineer

Offenburg, Germany

- Built a real-time ROS2 stack for a 7-DoF cobot (Kassow KR1410) integrating the robot's native API (KORD API), communicating at 250 Hz.
- Implemented a `ros2_control` hardware interface (`SystemInterface`) in C++, exposing position, velocity, and acceleration command interfaces and full digital I/O (8 outputs, 16 inputs), with lifecycle management.
- Ran the control thread at 250 Hz with `SCHED_FIFO` scheduling, `mlockall`, and CPU DMA latency pinning, using lock-free buffers to keep robot communication jitter-free.
- Using the API, validated and implemented joint-level position, velocity, acceleration, and torque control as standalone tests outside the ROS2 stack.
- Integrated MoveIt2 (OMPL, Pilz Industrial Motion Planner) for motion planning, configuring and tuning TOTG and Ruckig trajectory smoothing to generate time-optimal, jerk-limited joint trajectories.
- Developed a custom MoveIt post-processing plugin (`PlanningResponseAdapter`) that analyzes the linear and angular TCP speed of a planned trajectory against configurable safety limits before execution.
- Migrated the robot's control software from a legacy standalone script to the full ROS2/MoveIt2 architecture now used for ongoing development, replacing a single-purpose client with a modular, reusable motion stack.
- Built the pick-and-place application (80+ taught waypoints) with digital I/O controlling the gripper, synchronized with feedback from an analog distance sensor for object detection.
- Built a supervisor node that handles task priorities over ROS 2 topics, automatically routing the arm to a safe recovery pose on missed gripper grabs and skipping failed stations for alternate targets.

Institute for Advanced Biomechanics and Motion Studies

March 2026 – August 2026

Student Research Assistant - Software Engineer

Offenburg, Germany

- Contributed to designing a unified HDF5 schema for multi-modal sensor data (force plates, motion capture, IMU, EMG) with coordinate-frame calibration and transformation matrices, shared across research pipelines.
- Built Storage Scout (Python, PyQt6, SQLite), a desktop tool to audit and clean up research data across local, NAS, and network drives.
- Designed its concurrent scanning engine using producer-consumer thread queues, batched SQLite transactions, and WAL journaling to keep the UI responsive on large datasets.
- Set up local agentic AI dev environments (Codex, LM Studio) for privacy-conscious code generation and debugging workflows.

Boeing | International Space Station Program

Bengaluru, India

Software Engineer II

Jun 2025 – Aug 2025

- Built a cross-platform desktop tool (Python, Kivy) that packaged and extended migration scripts into an end-to-end workflow, cutting manual work by 99%.
- Parallelized migration processes on HPC infrastructure to accelerate large-scale data transfers.
- Developed Python-based validation and monitoring tools that ensured 100% data integrity during migration.

Software Engineer I

Aug 2023 – Jun 2025

- Automated MSSQL-to-MySQL database migration with Python scripts, cutting manual effort by 90%.
- Modernized a legacy VB6 application into a Python-based desktop application.
- Converted roughly 10,000 lines of T-SQL across 15 scripts into MySQL stored procedures.
- Managed scalable MySQL deployments on VMware Tanzu and Microsoft Azure with multi-instance support.

Education

Hochschule Offenburg <i>Master of Science in Communication and Media Engineering</i>	Oct 2025 – Present <i>Offenburg, Germany</i>
M S Ramaiah Institute of Technology <i>Bachelor of Engineering in Electronics and Instrumentation Engineering</i>	May 2019 – Jun 2023 <i>Bengaluru, India</i>

Projects

3-DOF Robotic Arm ROS 2 Simulation, Motion Planning & Voice Control	May 2026 – Sep 2026
- Built a full ROS 2 software stack for a 3-DoF robotic arm, modeling it in URDF/Xacro, implementing a ros2_control hardware interface, and configuring closed-loop joint control, validated in Gazebo/Ignition. - Integrated MoveIt 2 for collision-aware motion planning and built a task-based control system to manage pick and place, sleep, and wake behaviors of the robot. - Configured trajectory-based controllers for the arm and gripper with defined joint limits, enabling smooth, precise motion instead of raw position commands. - Added voice control by connecting an Alexa skill to the robot software, letting spoken commands trigger real-time arm movement.	
Automated Neonatal Jaundice Detection and Phototherapy Control System	Jan 2024– May 2024
- Built an end-to-end prototype combining computer vision, embedded control, and sensor feedback to automate neonatal phototherapy monitoring. - Implemented a real-time image-processing pipeline (OpenCV, colour-space analysis, contour detection) to estimate jaundice-related skin characteristics from camera images. - Built a closed-loop control system linking image analysis to an Arduino over serial, automatically switching the phototherapy lamp on or off based on detection results and safety conditions. - Added safety interlocks: automatic shutdown on overheating (via a temperature sensor) and automatic pause when the infant left the required treatment position. - Integrated SMS alerts (Twilio) and an on-device display to report treatment status, diagnostics, and safety events to caregivers in real time.	

Technical Skills

Robotics & Motion Planning: ROS 2 (Jazzy), ros2_control, MoveIt 2, OMPL, Pilz Industrial Motion Planner, URDF/Xacro, RViz2, Gazebo/Ignition and Linux Real-Time Kernel
Perception & Computer Vision: OpenCV, Image Processing, Colour-Space Analysis, Contour Detection
Embedded Systems & Hardware: Arduino, UART/Serial Communication, Digital I/O, Relay Control, Sensor Integration (IMU, Temperature, Distance Sensors)
Programming Languages: C++, Python, SQL/T-SQL
Software Engineering & Systems: PyQt6, SQLite, Multithreading/Concurrency, WAL Journaling, Flask, REST APIs, Kivy, HDF5, Git
Databases & Cloud: MySQL, MSSQL, VMware Tanzu, Microsoft Azure

Languages

English: C1
German: A2